

## Relationship between variables



1

- a Yes                      b No                      c No                      d Yes

2

- a No  
b Distance from the equator affects temperature.  
c No  
d Distance covered affects time of travel or time of travel affects distance covered.  
e No  
f Time spent working affects wages earned.  
g Number of pages affects time spent to finish a novel.  
h No  
i No  
j No  
k Number of pets affects time spent caring for pets.  
l Time spent training affects athletic performance.

3

- a  
i Temperature outside                      ii Number of ice cream cones sold  
b  
i Size of a pizza                      ii Cost of pizza  
c  
i Time spent studying                      ii Exam mark

4

- a Ticket sales affect the revenue made from a show.  
b Dependent variable - revenue from the show, Independent variable - number of ticket sales

5

- a The height of the kite and the speed of the wind  
b Dependent variable - height of the kite, Independent variable - speed of the wind

6

- a Number of soft drinks                      b Waist circumference  
c Number of soft drinks                      d Waist circumference

7 a Height of the ball



b Amount of time passed

8

- a There is no direct relationship between a student's mark in English and their mark in History.
- b As GDP per capita increases, infant mortality rate decreases.
- c As height increases, weight increases.

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## Scatter plots

9

a Amount of healthy coral

b Sea temperature

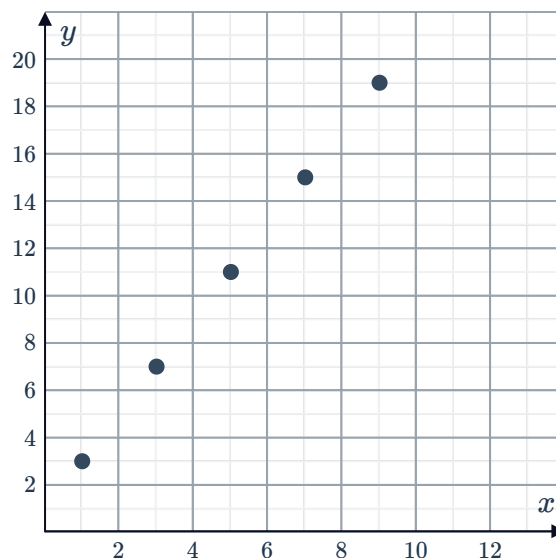
10

- a There is no direct relationship between someone's vision and their height.
- b As air temperature increases, sea temperature increases.

11

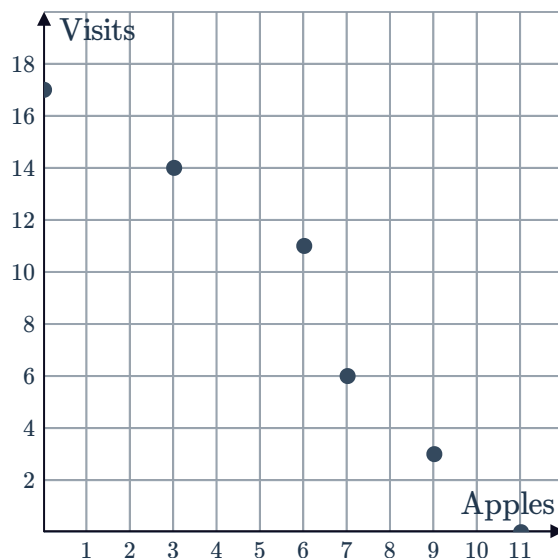
- a As land area increases, house price increases.
- b As temperature increases, soups sold decreases.
- c Height has no effect on salary.

12



13

a

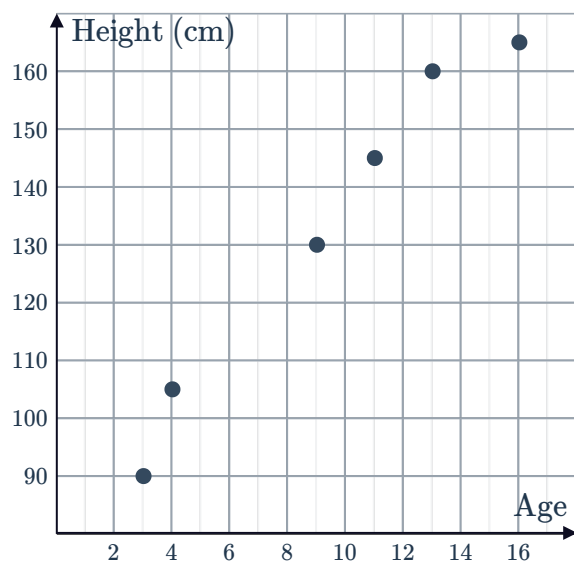


b 11

c No, because the data is only from six students and there are many other factors to consider.

14

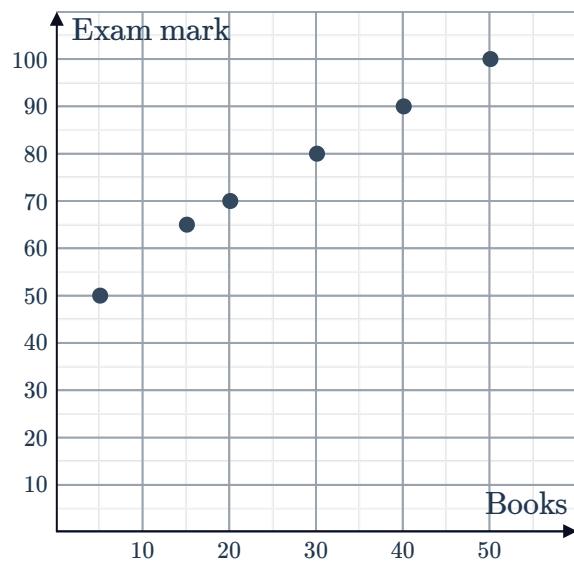
a



b 105 cm

15

a

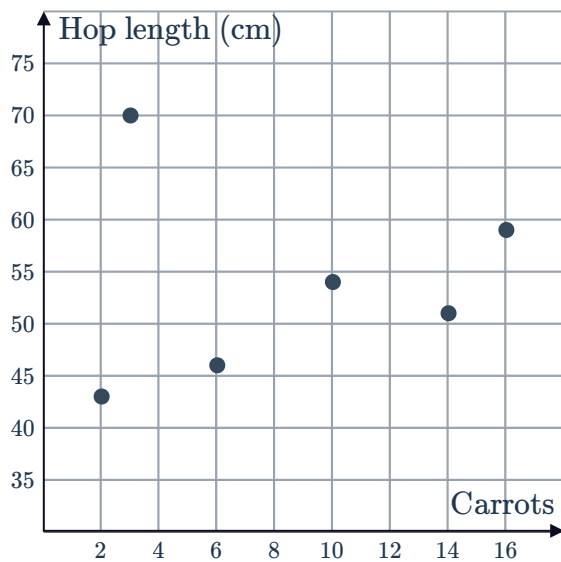


b 50

c Yes

16

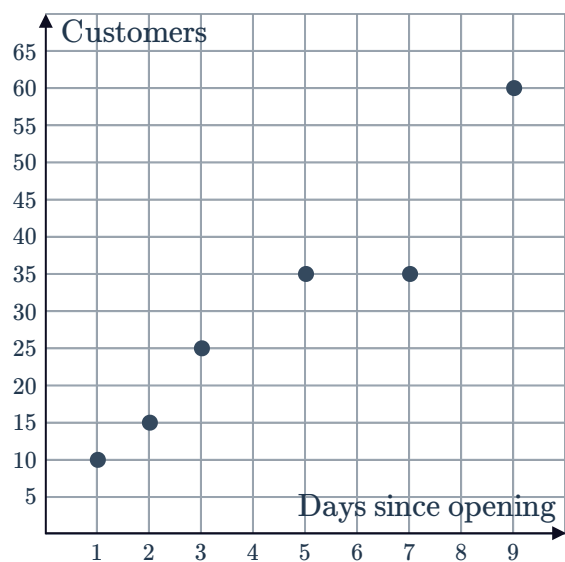
a



b 16 cm

17

a

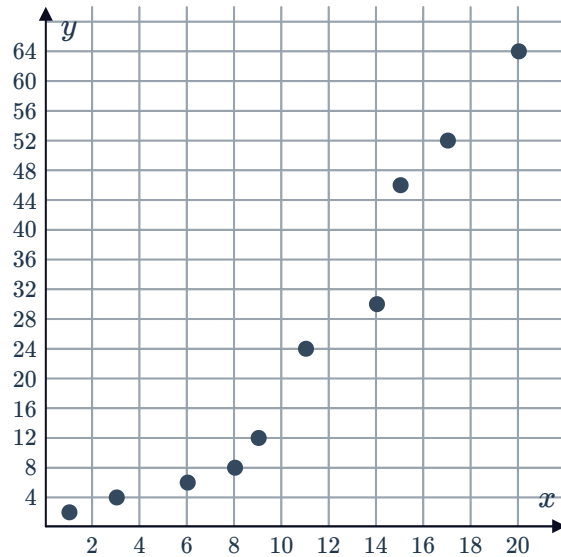


b

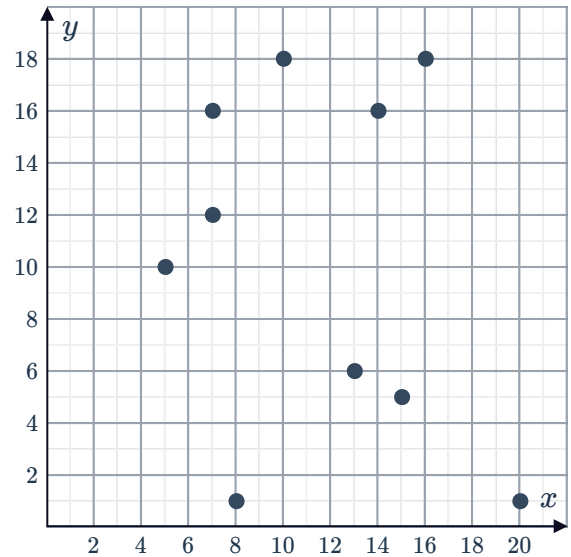
30



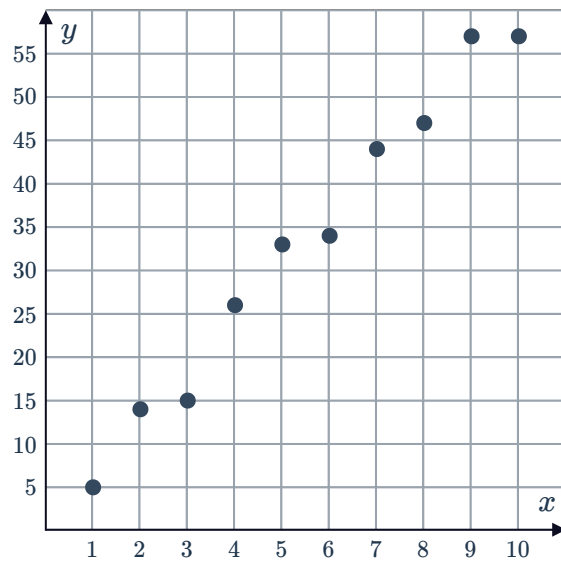
a



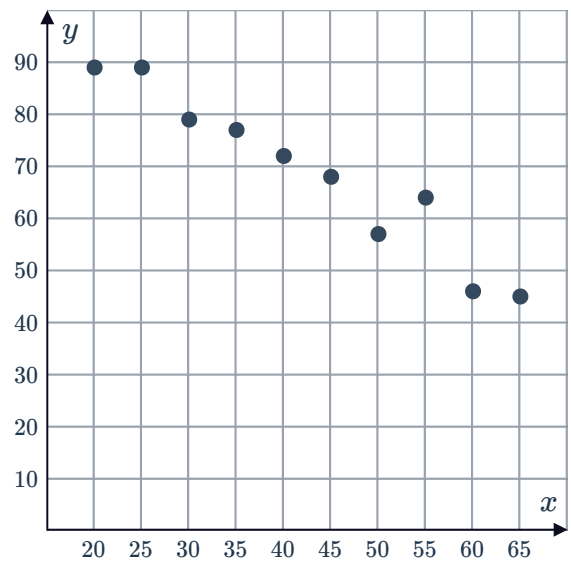
b



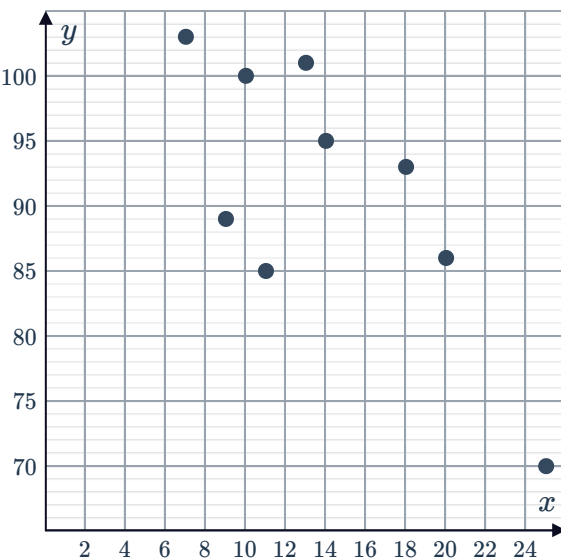
c



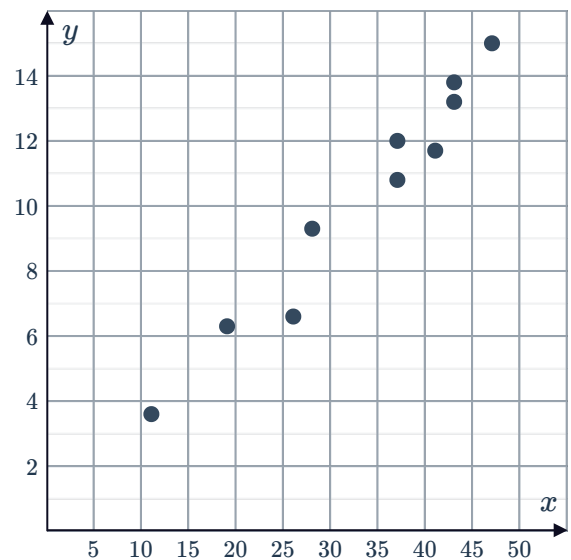
d



e



f

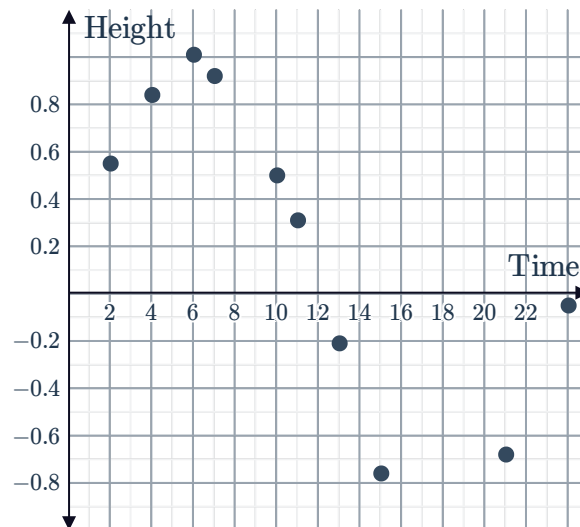


19



a Dependent variable - height, Independent variable - time

b



20 Dot 3

21

a The teacher

b Region 3

22

a Number of facebook checks

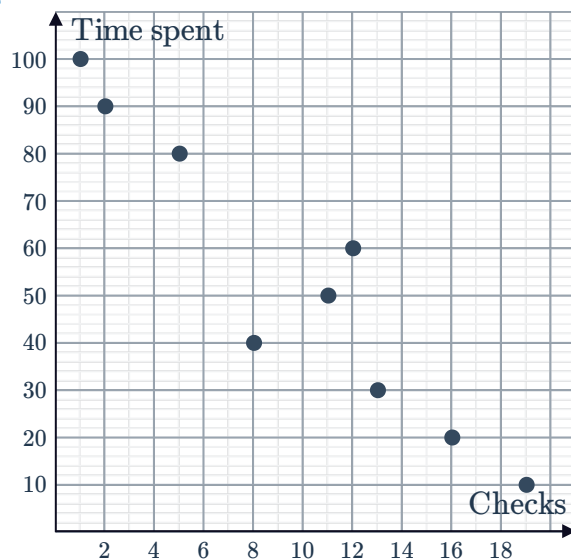
b Horizontal

c Time spent preparing for class

d Vertical

e

f Decreasing linear

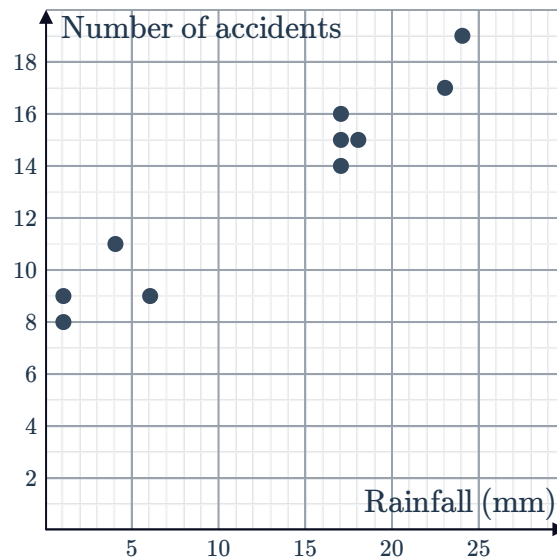


23



a Dependent variable - number of accidents, independent variable - amount of rainfall

b

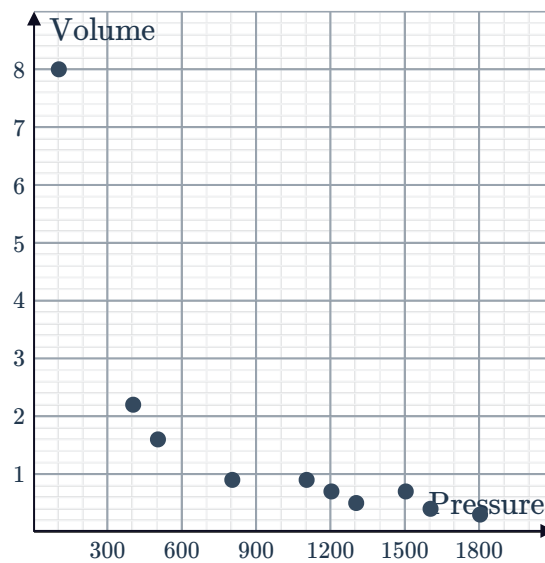


c As rainfall increases, number of accidents increases.

24

a Dependent variable - volume, Independent variable - pressure

b

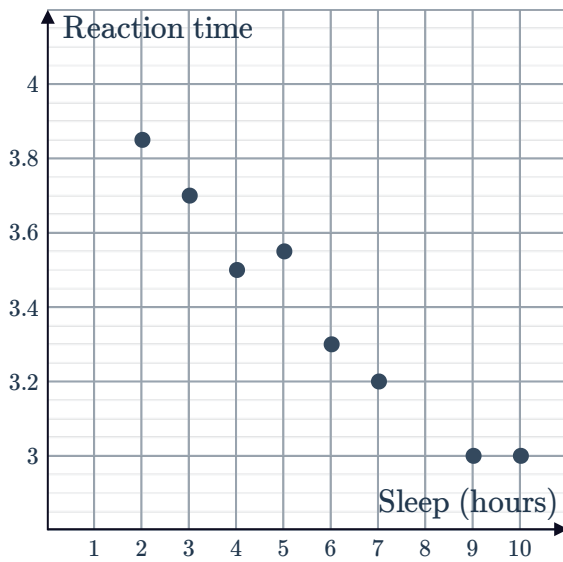


c As pressure increases, volume decreases.



25

a



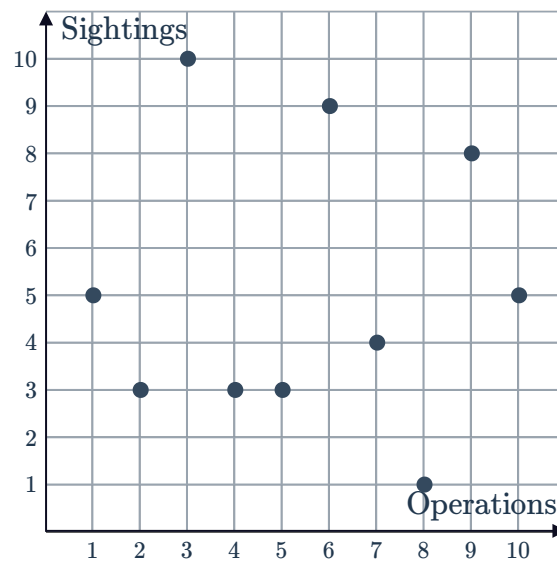
b Negatively

c Yes

26

a In that particular month, 1 shark was sighted and 8 cage diving operations took place.

b



c There is no definite relationship.

d No